



August 27, 2026 • Assignments

Assignment 1 Solutions – Introduction to Big Data

5 Marks Questions

1. Explain the evolution of Big Data and its importance in modern industries.

Big Data evolved as traditional systems (RDBMS, spreadsheets) failed to handle the massive volume of data generated by the internet, social media, sensors, and mobile devices from the early 2000s onward. It progressed through stages: structured data in relational databases, then semi-structured/unstructured data from web and social platforms, and now real-time streaming data from IoT devices. Technologies like Hadoop, Spark, and NoSQL databases emerged to store and process this scale of data.

It is important today because it helps industries make data-driven decisions, understand customer behavior, predict trends, optimize operations, detect fraud, and personalize services, giving businesses a competitive edge.

2. Differentiate between Structured, Semi-Structured, and Unstructured data with examples.

Type	Description	Example
Structured	Organized in fixed rows/columns, follows a schema	Excel sheets, SQL database tables
Semi-Structured	Has some organizational tags/markers but no fixed schema	JSON, XML, emails
Unstructured	No predefined format or structure	Images, videos, audio, social media posts

3. Explain the 3Vs of Big Data: Volume, Velocity, and Variety.

- **Volume:** The sheer amount of data generated, ranging from terabytes to petabytes, from sources like social media, transactions, and sensors.

- **Velocity:** The speed at which data is generated and needs to be processed, often in real time (e.g., stock market feeds, live sensor data).
- **Variety:** The different formats of data, structured, semi-structured, and unstructured, coming from multiple sources like text, images, and videos.

4. Compare Traditional Business Intelligence (BI) and Big Data Analytics.

Feature	Traditional BI	Big Data Analytics
Data type	Mostly structured	Structured, semi-structured, unstructured
Data volume	Limited/moderate	Massive (petabyte scale)
Processing	Batch processing, historical	Real-time and batch processing
Tools	SQL, data warehouses	Hadoop, Spark, NoSQL
Purpose	Reporting past performance	Predictive and prescriptive insights

5. Discuss the challenges faced in Big Data management and processing.

- **Storage:** Handling and storing massive volumes of data cost-effectively.
- **Processing speed:** Analyzing data quickly enough to be useful, especially real-time streams.
- **Data quality:** Dealing with incomplete, inconsistent, or duplicate data.
- **Security and privacy:** Protecting sensitive data from breaches and ensuring compliance.
- **Skill gap:** Shortage of professionals skilled in Big Data tools and analytics.
- **Integration:** Combining data from multiple, varied sources into a usable format.

4 Marks Questions

1. What is Big Data? Explain its characteristics.

Big Data refers to extremely large and complex datasets that traditional data processing tools cannot handle efficiently. Its main characteristics are the 5Vs:

- **Volume:** Huge amount of data
- **Velocity:** High speed of data generation
- **Variety:** Multiple data formats
- **Veracity:** Uncertainty/quality of data
- **Value:** Usefulness extracted from the data

2. Explain why Big Data is important for organizations.

Big Data helps organizations understand customer preferences, improve decision-making through data-driven insights, detect fraud and risks early, optimize supply chains and operations, and personalize marketing. It gives companies a competitive advantage by revealing patterns and trends that would otherwise go unnoticed.

3. Describe industry examples where Big Data is used effectively.

- **Healthcare:** Predicting disease outbreaks and personalizing treatment plans.
- **Retail:** Analyzing purchase patterns for recommendations (e.g., Amazon).
- **Banking:** Fraud detection and credit risk analysis.
- **Transportation:** Real-time traffic and route optimization (e.g., Uber, Ola).
- **Social Media:** Targeted advertising based on user behavior.

4. Define Big Data Analytics and explain its applications.

Big Data Analytics is the process of examining large and varied datasets to uncover hidden patterns, correlations, and insights that support better decision-making. Applications include fraud detection, customer segmentation, predictive maintenance, recommendation systems, and risk management across industries like finance, healthcare, and retail.

5. Explain the role of Data Science in Big Data.

Data Science provides the methods, algorithms, and tools (statistics, machine learning, programming) used to extract meaningful insights from Big Data. While Big Data deals with storing and managing massive datasets, Data Science analyzes that data to find patterns, build predictive models, and support decision-making.

2 Marks Questions

1. Define Structured Data.

Data organized in a fixed format with rows and columns, typically stored in relational databases (e.g., SQL tables).

2. What is Semi-Structured Data?

Data that does not follow a strict schema but contains tags or markers to separate elements, such as JSON or XML files.

3. Give two examples of Unstructured Data.

Images and videos; social media posts (also emails, audio files).

4. What is meant by Volume in Big Data?

Volume refers to the massive scale of data generated and stored, ranging from terabytes to petabytes.

5. Define Velocity in Big Data.

Velocity is the speed at which data is generated, collected, and processed, often in real time.

6. What is Variety in Big Data?

Variety refers to the different types and formats of data, structured, semi-structured, and unstructured, from diverse sources.

7. Define Big Data Analytics.

The process of examining large, complex datasets to discover patterns, trends, and insights for better decision-making.

8. What is Data Science?

An interdisciplinary field that uses statistics, programming, and domain knowledge to extract insights and knowledge from data.

9. State any two challenges of Big Data.

Data storage limitations and data security/privacy concerns (also: data quality issues, lack of skilled professionals).

10. Expand BI and explain its purpose.

BI stands for **Business Intelligence**. Its purpose is to collect, process, and analyze historical business data to generate reports and dashboards that help in monitoring performance and making informed decisions.

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